Gramática del analizador sintáctico de Lua 5.5

Notación

Los no terminales se escriben ⟨ESTE_FORMATO⟩ y los terminales en `máquina de escribir`. Los terminales son los tipos de token que produce el analizador léxico (`tkn_...`, `id`) y las palabras reservadas. El símbolo inicial es ⟨CHUNK⟩. ε es la cadena vacía, $\rightarrow$ introduce una regla y | separa sus alternativas. La gramática es LL(1).

1. Estructura del programa

Símbolo inicial, bloques y la sentencia `return`.

⟨CHUNK⟩ → ⟨BLOCK⟩

⟨BLOCK⟩ → ⟨STATLIST⟩ ⟨RETSTAT⟩

⟨STATLIST⟩ → ⟨STAT⟩ ⟨STATLIST⟩
| ε

⟨RETSTAT⟩ → `return` ⟨MAYBE_EXPLIST⟩ ⟨OPT_SEMILOCON⟩
| ε

⟨OPT_SEMILOCON⟩ → `tkn_semicolon`
| ε

2. Sentencias

Las asignaciones y las llamadas se factorizan después del primer nombre.

⟨STAT⟩ → `tkn_semicolon`
| ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨LABEL⟩
| `break`
| `goto` ⟨NAME⟩
| `while` ⟨EXP⟩ ⟨DO_BLOCK_END⟩
| `repeat` ⟨BLOCK⟩ `until` ⟨EXP⟩
| ⟨IFSTAT⟩
| ⟨FORSTAT⟩
| `function` ⟨FUNCNAME⟩ ⟨FUNCBODY⟩
| ⟨LOCAL_DEC⟩
| ⟨GLOBAL_DEC⟩

⟨VAR_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩
| `tkn_period` ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨FUNCTIONCALL⟩ ⟨FUNCALL_STAT⟩
| ⟨MORE_VAR⟩ `tkn_assign` ⟨EXPLIST⟩

⟨FUNCALL_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩
| `tkn_period` ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨FUNCTIONCALL⟩ ⟨FUNCALL_STAT⟩
| ε

⟨LABEL⟩ → `tkn_goto` ⟨NAME⟩ `tkn_goto`

⟨DO_BLOCK_END⟩ → `do` ⟨BLOCK⟩ `end`

⟨IFSTAT⟩ → `if` ⟨EXP⟩ `then` ⟨BLOCK⟩ ⟨ELSEIFSTAT⟩ ⟨ELSESTAT⟩ `end`

⟨ELSEIFSTAT⟩ → `elseif` ⟨EXP⟩ `then` ⟨BLOCK⟩ ⟨ELSEIFSTAT⟩
| ε

⟨ELSESTAT⟩ → `else` ⟨BLOCK⟩
| ε

⟨FORSTAT⟩ → `for` ⟨NAME⟩ ⟨FORTYPE⟩ ⟨DO_BLOCK_END⟩

⟨FORTYPE⟩ → ⟨NUMERICAL_FOR⟩
| ⟨GENERIC_FOR⟩

⟨NUMERICAL_FOR⟩ → `tkn_assign` ⟨EXP⟩ `tkn_comma` ⟨EXP⟩ ⟨OPT_EXP⟩

⟨OPT_EXP⟩ → `tkn_comma` ⟨EXP⟩
| ε

⟨GENERIC_FOR⟩ → ⟨MORE_NAMES⟩ `in` ⟨EXPLIST⟩

3. Declaraciones

`local` y `global`.

⟨LOCAL_DEC⟩ → `local` ⟨LOCAL_OPTS⟩

⟨LOCAL_OPTS⟩ → ⟨FUNC_DEC⟩
| ⟨ATTNAME_BRANCH⟩

⟨ATTNAME_BRANCH⟩ → ⟨ATTNAMELIST⟩ ⟨OPT_EXPLIST⟩

⟨GLOBAL_DEC⟩ → `global` ⟨GLOBAL_OPTS⟩

⟨GLOBAL_OPTS⟩ → ⟨FUNC_DEC⟩
| ⟨ATTNAMELIST⟩ ⟨OPT_EXPLIST⟩

⟨FUNC_DEC⟩ → `function` ⟨NAME⟩ ⟨FUNCBODY⟩

⟨OPT_EXPLIST⟩ → `tkn_assign` ⟨EXPLIST⟩
| ε

⟨ATTNAMELIST⟩ → ⟨NAME⟩ ⟨ATTRIB⟩ ⟨NAME_ATTRIB_LIST⟩

⟨NAME_ATTRIB_LIST⟩ → `tkn_comma` ⟨NAME⟩ ⟨ATTRIB⟩ ⟨NAME_ATTRIB_LIST⟩
| ε

⟨ATTRIB⟩ → `tkn_less` ⟨NAME⟩ `tkn_greater`
| ε

4. Funciones

Nombres de función, cuerpos y listas de parámetros.

⟨FUNCNAME⟩ → ⟨NAME⟩ ⟨DOTTED_FUNCNAME⟩ ⟨COLON_FUNCNAME⟩

⟨DOTTED_FUNCNAME⟩ → `tkn_period` ⟨NAME⟩ ⟨DOTTED_FUNCNAME⟩
| ε

⟨COLON_FUNCNAME⟩ → `tkn_colon` ⟨NAME⟩
| ε

⟨FUNCTIONDEF⟩ → `function` ⟨FUNCBODY⟩

⟨FUNCBODY⟩ → `tkn_opening_par` ⟨OPT_PARLIST⟩ `tkn_closing_par` ⟨BLOCK⟩ `end`

⟨OPT_PARLIST⟩ → ⟨PARLIST⟩
| ε

⟨PARLIST⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩
| ⟨VARPARAGRAM⟩

⟨PARLIST_TAIL⟩ → `tkn_comma` ⟨PARLIST_AFTER_COMMA⟩
| ε

⟨PARLIST_AFTER_COMMA⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩
| ⟨VARPARAGRAM⟩

⟨VARPARAGRAM⟩ → `tkn_varargs` ⟨OPT_NAME⟩

⟨OPT_NAME⟩ → ⟨NAME⟩
| ε

5. Listas

Listas de variables, de nombres y de expresiones.

⟨MORE_NAMES⟩ → `tkn_comma` ⟨NAME⟩ ⟨MORE_NAMES⟩
| ε

⟨MORE_VAR⟩ → `tkn_comma` ⟨PREFIXEXP⟩ ⟨MORE_VAR⟩
| ε

⟨MAYBE_EXPLIST⟩ → ⟨EXPLIST⟩
| ε

⟨EXPLIST⟩ → ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩

⟨MORE_EXP⟩ → `tkn_comma` ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩
| ε

⟨EXPLIST_ITEM⟩ → ⟨EXP⟩
| ⟨FUNCTIONDEF⟩

6. Expresiones

Operandos, expresiones prefijo y llamadas.

⟨EXP⟩ → ⟨OPERAND⟩ ⟨EXP_PRIME⟩

⟨EXP_PRIME⟩ → ⟨ARITH_TAIL⟩ ⟨CMP_TAIL⟩ ⟨LOGIC_TAIL⟩

⟨OPERAND⟩ → ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩
| ⟨OPERAND_NOT_ID⟩

⟨OPERAND_NOT_ID⟩ → `nil`
| `false`
| `true`

$$
\begin{aligned}
&\mid \texttt{tkn_num} \\
&\mid \texttt{tkn_str} \\
&\mid \langle \text{TABLECONSTRUCTOR} \rangle \\
&\mid \langle \text{UNOP} \rangle \ \langle \text{OPERAND} \rangle \\
&\mid \texttt{tkn_opening_par} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_par} \\
\langle \text{PREFIXEXP} \rangle &\rightarrow \langle \text{NAME} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
\langle \text{PREFIXEXP_PRIME} \rangle &\rightarrow \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \texttt{tkn_period} \ \langle \text{NAME} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \langle \text{FUNCTIONCALL} \rangle \ \langle \text{PREFIXEXP_PRIME} \rangle \\
&\mid \varepsilon \\
\langle \text{FUNCTIONCALL} \rangle &\rightarrow \langle \text{ARGS} \rangle \\
&\mid \texttt{tkn_colon} \ \langle \text{NAME} \rangle \ \langle \text{ARGS} \rangle \\
\langle \text{ARGS} \rangle &\rightarrow \texttt{tkn_opening_par} \ \langle \text{MAYBE_EXPLIST} \rangle \ \texttt{tkn_closing_par}
\end{aligned}
$$

7. Tablas

Constructores de tablas.

$$
\begin{aligned}
\langle \text{TABLECONSTRUCTOR} \rangle &\rightarrow \texttt{tkn_opening_key} \ \langle \text{FIELDLIST} \rangle \ \texttt{tkn_closing_key} \\
\langle \text{FIELDLIST} \rangle &\rightarrow \langle \text{FIELD} \rangle \ \langle \text{FIELD_TAIL} \rangle \\
&\mid \varepsilon \\
\langle \text{FIELD_TAIL} \rangle &\rightarrow \langle \text{FIELDSEP} \rangle \ \langle \text{FIELDLIST} \rangle \\
&\mid \varepsilon \\
\langle \text{FIELD} \rangle &\rightarrow \texttt{tkn_opening_bra} \ \langle \text{EXP} \rangle \ \texttt{tkn_closing_bra} \ \texttt{tkn_assign} \ \langle \text{FIELD_VALUE} \rangle \\
&\mid \langle \text{NAME} \rangle \ \langle \text{FIELD_ID} \rangle \\
&\mid \langle \text{OPERAND_NOT_ID} \rangle \ \langle \text{EXP_PRIME} \rangle \\
&\mid \langle \text{FUNCTIONDEF} \rangle \\
\langle \text{FIELD_ID} \rangle &\rightarrow \texttt{tkn_assign} \ \langle \text{FIELD_VALUE} \rangle \\
&\mid \langle \text{PREFIXEXP_PRIME} \rangle \ \langle \text{EXP_PRIME} \rangle \\
\langle \text{FIELD_VALUE} \rangle &\rightarrow \langle \text{EXP} \rangle \\
&\mid \langle \text{FUNCTIONDEF} \rangle \\
\langle \text{FIELDSEP} \rangle &\rightarrow \texttt{tkn_comma} \\
&\mid \texttt{tkn_semicolon}
\end{aligned}
$$

8. Operadores

Operadores binarios por nivel, operadores unarios y nombres.

$$
\begin{aligned}
\langle \text{LOGIC_TAIL} \rangle &\rightarrow \langle \text{LOGICAL_OP} \rangle \ \langle \text{CMP_EXP} \rangle \ \langle \text{LOGIC_TAIL} \rangle \\
&\mid \varepsilon \\
\langle \text{CMP_EXP} \rangle &\rightarrow \langle \text{ARTH_EXP} \rangle \ \langle \text{CMP_TAIL} \rangle \\
\langle \text{CMP_TAIL} \rangle &\rightarrow \langle \text{RELATIONAL_OP} \rangle \ \langle \text{ARTH_EXP} \rangle \\
&\mid \varepsilon \\
\langle \text{ARTH_EXP} \rangle &\rightarrow \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \\
\langle \text{ARITH_TAIL} \rangle &\rightarrow \langle \text{FLAT_OP} \rangle \ \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \\
&\mid \varepsilon
\end{aligned}
$$

⟨FLAT_OP⟩ → ⟨ARITH_OP⟩
| ⟨BITWISE_OP⟩
| ⟨OTHER_OP⟩

⟨ARITH_OP⟩ → `tkn_plus`
| `tkn_minus`
| `tkn_times`
| `tkn_div`
| `tkn_floor_div`
| `tkn_power`
| `tkn_mod`

⟨RELATIONAL_OP⟩ → `tkn_less`
| `tkn_leq`
| `tkn_greater`
| `tkn_geq`
| `tkn_equal`
| `tkn_neq`

⟨LOGICAL_OP⟩ → `and`
| `or`

⟨BITWISE_OP⟩ → `tkn_bit_and`
| `tkn_bitex_or`
| `tkn_bit_or`
| `tkn_right_shift`
| `tkn_left_shift`

⟨OTHER_OP⟩ → `tkn_concat`

⟨UNOP⟩ → `tkn_minus`
| `not`
| `tkn_length`

⟨NAME⟩ → `id`
| `print`
| `pcall`
| `error`
| `warn`
| `xpcall`